

Q3 Inventory Service Review

This review assesses the inventory-service fixture code ahead of the Q3 release. The service is healthy overall, but three bugs need fixing before ship; recommendations and a bug summary follow.

Current State

Three endpoints are working and eight tests are passing.

Issues Found

bulk_restock has an off-by-one index pattern that rebuilds the key list on every iteration, making the operation quadratic.

fulfill_by_skus has an empty-order semantic bug that bypasses availability checks when an order is empty.

average_stock risks a ZeroDivisionError on an empty state, and its docstring claims zero-stock items are excluded when they are not.

Recommendations

Refactor bulk_restock to iterate with .items().

Validate empty orders before fulfillment.

Guard average_stock against an empty state.

Bug Summary

Bug	Location	Severity
Off-by-one index pattern	bulk_restock	Important
Empty-order semantic bug	fulfill_by_skus	Important
ZeroDivisionError risk	average_stock	Important